



YMDS AI

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Manual policy review is slow, repetitive, and legally risky.

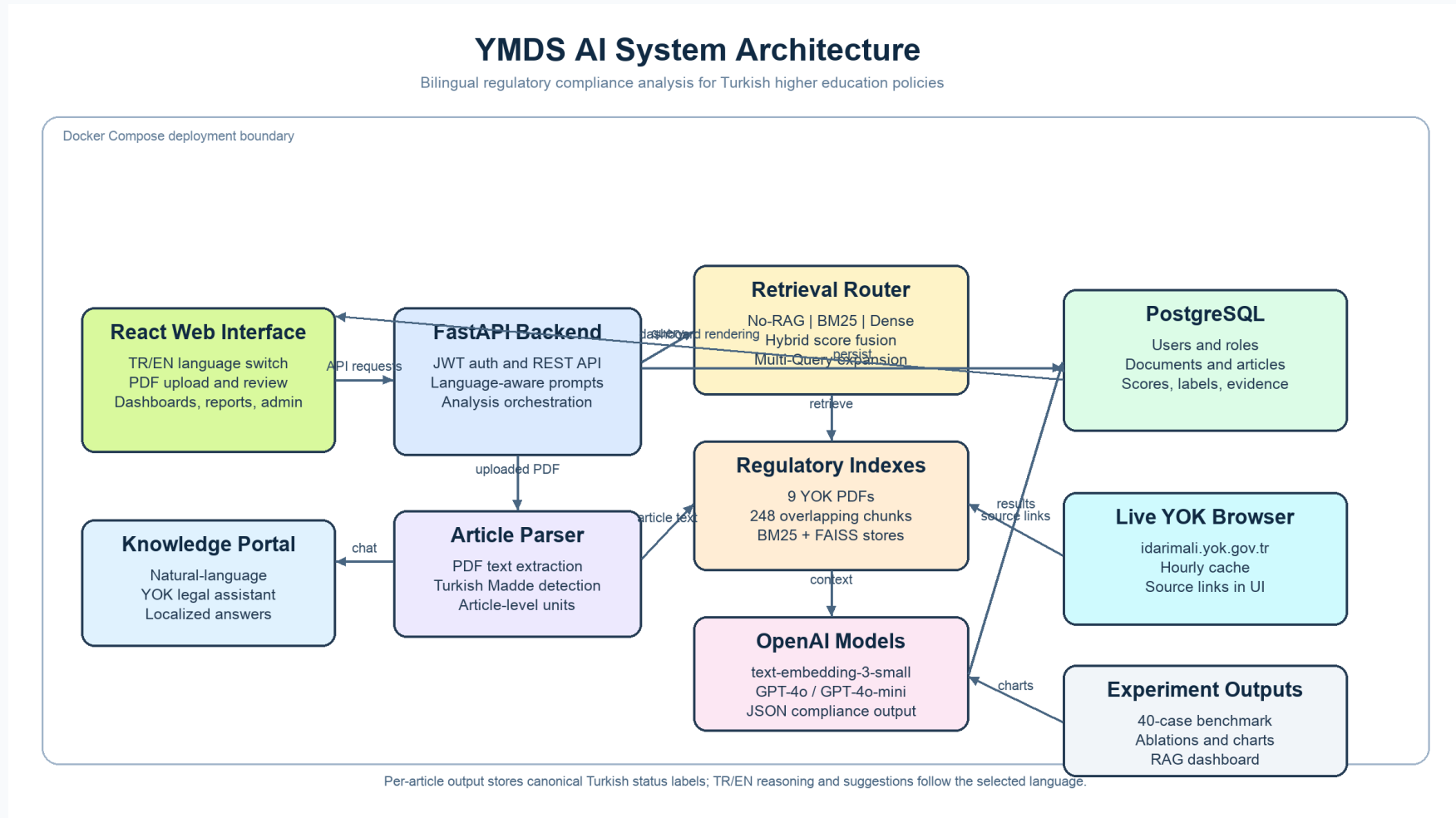
- Universities must align internal regulations with YOK legislation
- Manual comparison across long legal PDFs is error-prone
- Administrators need explainable article-level evidence



RAG turns policy review into a grounded retrieval and reasoning workflow.

- Parse uploaded PDFs into articles
- Retrieve relevant YOK passages with BM25, Dense, Hybrid, or Multi-Query
- Use GPT-4o / GPT-4o-mini to produce structured compliance JSON
- Keep human reviewers in the loop

Dockerized React + FastAPI + PostgreSQL deployment.





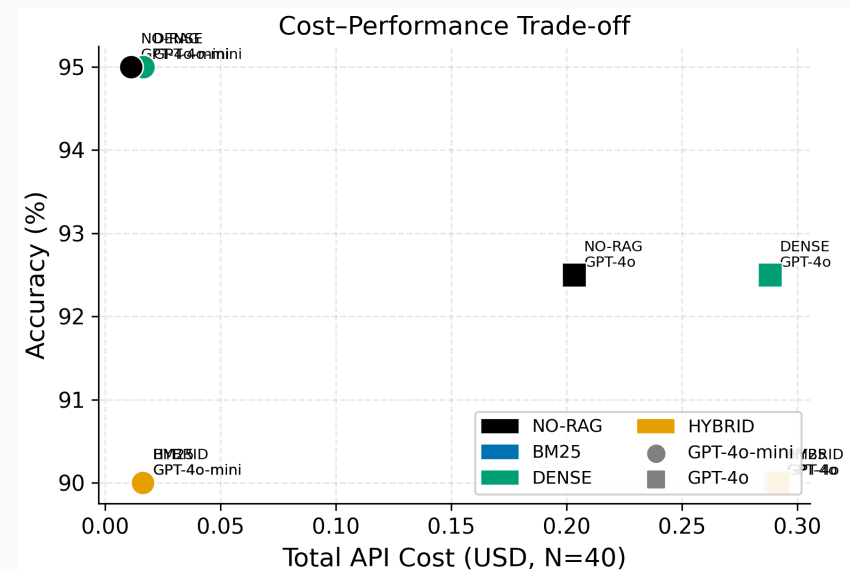
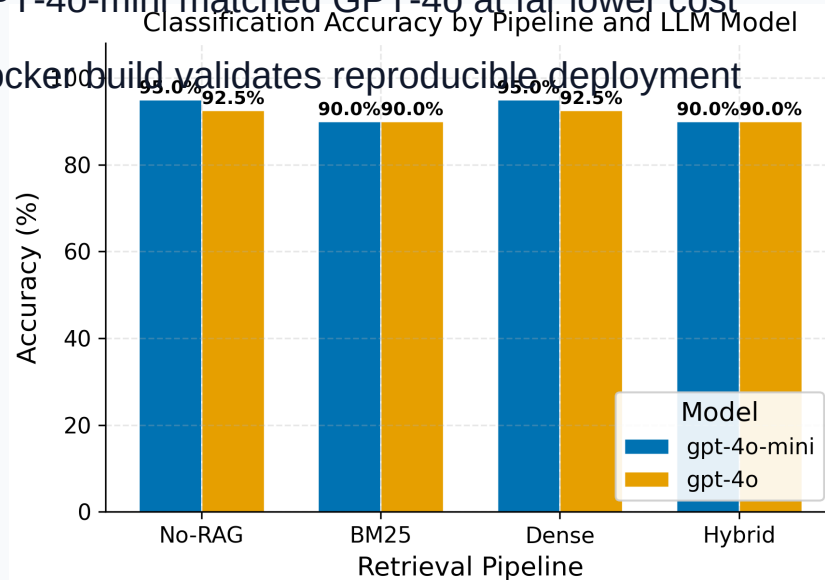
Senior Design II converted the prototype into a working research platform.

- Built article-level parser and RAG service
- Added admin dashboard, reports, archive, and YOK browser
- Benchmarked five retrieval strategies
- Added TR/EN interface and localized AI outputs

Results & Testing I

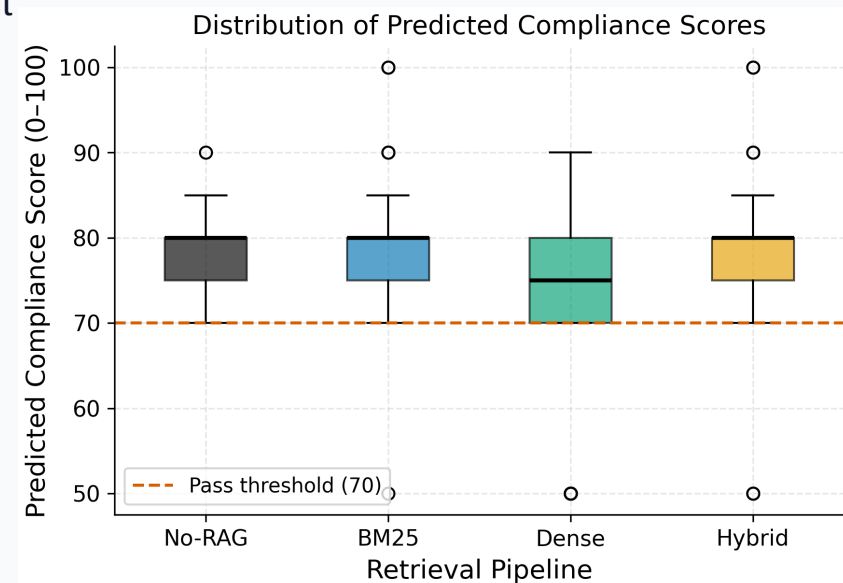
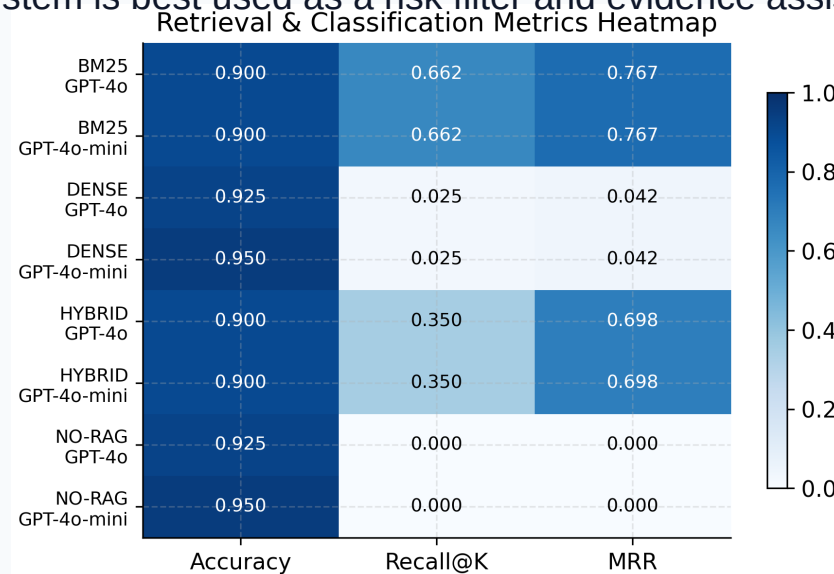
RAG improves over direct prompting.

- 40-case RAG-aware benchmark
- Hybrid and BM25 reached the best main accuracy group
- GPT-4o-mini matched GPT-4o at far lower cost
- Docker build validates reproducible deployment



The strongest limitation is non-compliant recall.

- Models tend to hedge toward partial compliance
- Human legal review remains mandatory
- System is best used as a risk filter and evidence assistant





A 60-second live walkthrough for the presentation.

- Login and select language
- Upload a policy PDF
- Choose Hybrid + GPT-4o-mini
- Inspect article evidence, reasoning, and suggestions
- Ask a follow-up question in Knowledge Portal



YMDS AI demonstrates practical AI-assisted regulatory review.

- End-to-end web application is functional in Docker
- RAG grounding improves traceability and usability
- Thanks to Istanbul Bilgi University and our advisor



Next steps for reliability and institutional deployment.

- Automatic regulation re-indexing
- Larger expert-labeled benchmark
- Calibration for non-compliant cases
- Audit logs and role-based export workflows



10-minute timing plan.

- Asya: problem, implementation, demo, UI and deployment
- Utku: methodology, experiments, results, limitations
- Both: Q&A and future work